

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	Slot-A	Date:	

Code of Conduct

I J.Raju(192325117) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: J.Raju

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:
Incorrect interpretation of ambiguous sentences

Failure to handle subject–verb agreement
Inefficient parsing for long queries Example
user query:

“Show me the transactions with the card from last month”

- Analyze the ambiguity present in the given sentence using CFG
- Evaluate the limitations of CFG in handling real-world conversational queries
- Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
- Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues

Poor handling of incomplete or ambiguous input

Difficulty in real-time response Example
command:

“Book a flight to Delhi with a window seat”

- Analyze how the given sentence can lead to multiple parse structures
- Analyze the limitations of top-down parsing in this real-time application
- Analyze how Earley parsing can improve handling of ambiguity and partial inputs
- Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:

Complex and lengthy medical sentences

Ambiguity in sentence structure

Accurate subject-verb agreement

Relationships between medical actions and entities (sub-categorization)

Real-time processing for hospital use Example input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.

- Design a complete NLP architecture integrating:

CFG for syntactic structure

Probabilistic CFG (PCFG) for ambiguity resolution

Feature Structures for agreement handling

Efficient parsing techniques

- Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).
- Create a method to:

Incorporate sub-categorization frames for medical verbs

Handle real-time data processing

Improve accuracy and scalability in a hospital environment

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

1. AI-Powered Banking Customer Support Chatbot

Example Query: “Show me the transactions with the card from last month.”

a) Ambiguity Using CFG

A simple CFG may contain rules such as:

```
S → VP
VP → V NP
NP → Det N
NP → NP PP
PP → P NP
```

The sentence contains prepositional-phrase attachment ambiguity. The phrase “with the card” and the phrase “from last month” can be attached to different parts of the sentence.

Interpretation 1: Show transactions that were made with the card and occurred last month.

```
Show → transactions → with the card + from last month
```

Interpretation 2: “from last month” may be attached to the card phrase rather than directly to transactions.

```
Show → transactions → with the card → from last month
```

A basic CFG can generate multiple valid structures but cannot determine which interpretation is most likely.

b) Limitations of CFG

Limitation	Effect
Ambiguity	May generate several valid parse trees without ranking them.
No probability	Cannot automatically choose the most likely interpretation.
Agreement	Basic CFG does not efficiently enforce subject–verb agreement.
Long queries	Possible parse structures can grow significantly.
Conversational input	Users may use incomplete or grammatically incorrect queries.
Context	CFG does not naturally use previous conversation context.
Domain knowledge	CFG does not know banking relationships such as card → transaction.

c) Improved Solution: PCFG + Feature Structures + Earley Parsing

PCFG: A Probabilistic Context-Free Grammar assigns probabilities to grammar rules. For example, $VP \rightarrow V NP$ [0.65] and $VP \rightarrow V VP$ [0.35]. The system can select the highest-probability parse.

Feature Structures: Features such as NUMBER, PERSON, and TENSE can represent grammatical information and use unification to prevent invalid subject–verb combinations.

Earley Parser: Earley parsing is a chart-parsing algorithm that maintains possible parsing states and is effective for ambiguous CFGs, recursive structures, and partial inputs.

d) Industry Benefits

The combined approach improves accuracy because PCFG ranks competing interpretations, feature structures enforce grammatical constraints, and Earley parsing efficiently handles complex and ambiguous queries. A safe structured interpretation is:

```
INTENT = SHOW_TRANSACTIONS
PAYMENT_METHOD = CARD
TIME_PERIOD = LAST_MONTH
```

This allows the banking chatbot to retrieve the appropriate transactions, reduce misunderstanding, and provide scalable customer support.

2. Voice Assistant Parsing System

Example Command: “Book a flight to Delhi with a window seat.”

a) Multiple Parse Structures

Parse 1 – intended interpretation:

Book → a flight → to Delhi + with a window seat Meaning:
Book a flight to Delhi and request a window seat.

Parse 2 – alternative attachment:

Book → a flight → to Delhi → with a window seat
A basic grammar may produce both structures without knowing which one matches the user's intent.

b) Limitations of Top-Down Parsing

- Backtracking: an incorrect grammar choice may require returning and trying another rule.
- Ambiguous input: multiple possibilities can cause repeated parsing attempts.
- Incomplete commands: partial speech such as “Book a flight to...” may cause failure.
- Left recursion: naive implementations can cause repeated or infinite expansion.
- Real-time latency: extensive backtracking can slow voice responses.

c) Earley Parsing for Ambiguity and Partial Input

Earley parsing uses a chart and three main operations: Predictor, Scanner, and Completer. It can maintain multiple interpretations simultaneously and preserve valid partial analyses as words arrive incrementally. This is useful for streaming speech recognition.

For the command, the desired semantic structure is:

INTENT = BOOK_FLIGHT
DESTINATION = DELHI
SEAT_PREFERENCE = WINDOW

d) Top-Down vs Earley

Feature	Top-Down Parsing	Earley Parsing
Approach	Starts from grammar start symbol	Uses a dynamic chart
Backtracking	Often required	Greatly reduced
Ambiguity	Poor/moderate	Good
Partial input	Poor	Better
Left recursion	Problematic in naive versions	Supported
Repeated computation	Can occur	Previous states are stored
Complex CFG	Less suitable	Suitable
Voice assistant use	Limited	Better choice

For a real-time voice assistant, Earley parsing is preferable because it handles ambiguity, recursion, and incremental input more effectively.

3. Healthcare NLP System

Example Input: “The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

a) Complete NLP Architecture

A suitable architecture combines CFG, PCFG, Feature Structures, medical NER, sub-categorization, semantic role analysis, and efficient parsing.

Medical Report → Text Preprocessing → Tokenization & Sentence Splitting → Medical NER → CFG → Feature Structures → PCFG → Earley/Chart Parser → Sub-categorization Analysis → Semantic Role

for Syntactic Structure

CFG identifies the basic grammatical structure. Example rules include:

```
S → NP VP
NP → Det N
NP → NP RelativeClause
VP → V NP
VP → V VP
```

The system identifies “The doctor” as the subject, “who reviewed the patient last week” as a relative clause, “recommends” as the main verb, and the two coordinated actions as starting medication and scheduling a follow-up visit.

PCFG for Ambiguity Resolution

PCFG assigns probabilities to alternative parse trees and selects the most likely structure. Probabilities can be learned from annotated medical corpora. **Feature Structures for Agreement**

Example:

```
doctor: NUMBER = Singular, PERSON = Third
recommends: NUMBER = Singular, PERSON = Third, TENSE = Present The
```

features match, so “Doctor recommends” is grammatically valid.

Efficient Parsing

An Earley or chart parser stores intermediate results, reducing unnecessary repeated computation and supporting ambiguous or recursive medical sentences.

b) Step-by-Step Workflow

- 1. Input Processing:** Receive the medical report.
- 2. Tokenization:** Split the sentence into words/tokens.
- 3. POS Tagging:** Identify nouns, verbs, proper nouns, and other grammatical categories.
- 4. Medical NER:** Identify Doctor, Patient, Medication, Follow-up Visit, and Chennai.
- 5. CFG Parsing:** Build the syntactic structure and identify the relative clause and coordinated actions.
- 6. PCFG Disambiguation:** Rank possible parses and select the most probable medical interpretation.
- 7. Feature Checking:** Verify subject–verb agreement such as doctor → recommends.
- 8. Semantic Role Analysis:** Determine agent, patient, treatment, follow-up action, time, and location.
- 9. Structured Output:** Store extracted information in a machine-readable clinical structure.

Structured Output

Information	Extracted Value
Diagnosis	Not explicitly stated
Agent	Doctor
Patient	Patient
Previous Action	Reviewed patient
Time	Last week
Recommended Treatment	Start medication
Information	Extracted Value

Follow-up Action	Schedule follow-up visit
Location	Chennai

Important: the sentence does not state a diagnosis. A reliable healthcare NLP system should return “not explicitly stated” rather than inventing a diagnosis.

c) Sub-Categorization, Real-Time Processing and Scalability

Sub-categorization frames:

```
review = Agent + Patient recommend =
Agent + Recommended_Action start =
Treatment
schedule = Clinical_Event + optional Location Applied
```

to the sentence:

```
RECOMMEND(Agent = Doctor, Action1 = START(Medication), Action2 = SCHEDULE(FollowUpVisit, Chennai))
```

Real-Time Processing:

```
Medical Report → Fast Preprocessing → NER + POS Tagging → Earley/Chart Parsing → PCFG Ranking →
Feature Unification → Medical Semantic Extraction → Structured Record → Hospital Information System
```

Accuracy and Scalability:

- Use CFG for grammatical structure.
- Use PCFG for probability-based ambiguity resolution.
- Use feature structures for grammatical constraints.
- Use Earley/chart parsing for efficient ambiguous parsing.
- Use medical NER for diseases, drugs, treatments, procedures, and locations.
- Use sub-categorization frames for medical verb–argument relationships.
- Use medical ontologies and domain terminology.
- Use confidence scores to flag uncertain outputs for human review.
- Use parallel or distributed processing for large hospital workloads.

Final Architecture

```
MEDICAL REPORT → PREPROCESSING → TOKENIZATION + POS → MEDICAL NER → CFG + FEATURE STRUCTURES →
PCFG → EARLEY/CHART PARSER → SUB-CATEGORIZATION → SEMANTIC ROLE ANALYSIS → MEDICAL INFORMATION
EXTRACTION → STRUCTURED PATIENT RECORD
```

Conclusion

The hybrid architecture is more suitable than basic CFG alone because it combines syntactic rules, probability-based ambiguity resolution, grammatical feature checking, efficient parsing, and medical-domain knowledge. It provides more accurate and scalable extraction of clinical information while supporting real-time hospital information systems.